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TIMERS



This chapter is about timers. A *timer* allows your program to count or wait for a given amount of time before moving on to perform some other action. In the world of text-based Python programs, this is easily achieved with `time.sleep()` by specifying a number of seconds to sleep. To pause for two and a half seconds, you could write:

```
import time
time.sleep(2.5)
```

However, in the world of pygame, and event-driven programming in general, the user should always be able to interact with the program, so pausing in this way is inappropriate. A call to `time.sleep()` would make the program nonreactive during the sleeping period.

Instead, the main loop needs to continue to run at whatever frame rate you have chosen. You need a way for the program to continue to loop, but also count time from a given starting point to some time in the future. There are three different ways this can be accomplished:

- Measure time by counting frames.
- Use pygame to create an event that is issued in the future.